

David Tong Standard Model Solutions — Sheet 3: Anomalies

D. Tong, The Standard Model, Example Sheet 3; 6 worked problems.

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Quantum Field Theory

Contents

1	Sheet 3, Problem 1: Chiral $U(1)$ Charge Assignments	1
2	Sheet 3, Problem 2*: Gauge and 't Hooft Anomalies in a Chiral Theory	1
3	Sheet 3, Problem 3*: Real Hypercharges and Anomaly Cancellation	3
4	Sheet 3, Problem 4: Integer Hypercharges	3
5	Sheet 3, Problem 5: Global Symmetries of the Standard Model	4
6	Sheet 3, Problem 6: Surviving Discrete Symmetries	5

These prompts paraphrase David Tong's The Standard Model, Example Sheet 3 (February 2026). Problem numbers, starred status, equations, and guidance follow the official sheet.

1. Sheet 3, Problem 1: Chiral $U(1)$ Charge Assignments

Prove that four left-handed Weyl fermions with integer charges cannot form a consistent chiral $U(1)$ gauge theory that can also couple to gravity. Exhibit a consistent example with five such fermions.

Solution. There are two constraints in this case,

$$\sum_{\text{left}} q_i^3 = \sum_{\text{right}} q_i^3, \quad \sum q_i = 0. \quad (1.1)$$

In the case of four Weyl spinors, we must either have one or two negative charges, namely $\{x, y, z, -w\}$ or $\{x, y, -z, -w\}$ where $x, y, z, w \in \mathbb{Z}^+$. In the first case, we have two equations

$$x^3 + y^3 + z^3 = w^3, \quad x + y + z = w, \quad (1.2)$$

which becomes

$$x^2y + xy^2 + y^2z + yz^2 + z^2x + zx^2 = 0, \quad (1.3)$$

which only has the trivial solution $x = y = z = w = 0$. The same argument works for the second case. Therefore, there cannot be any consistent chiral gauge theory with four left-handed spinors.

For five left-handed Weyl fermions, $\{1, 5, -7, -8, 9\}$ does the job.

2. Sheet 3, Problem 2*: Gauge and 't Hooft Anomalies in a Chiral Theory

Couple $SU(N)$ to one left-handed Weyl fermion χ in the two-index symmetric representation and to p left-handed Weyl fermions ψ_i in the anti-fundamental. Determine p for gauge anomaly cancellation.

Classically, $SU(p)$ rotates the ψ_i , while $U(1)_\chi$ rotates only χ and $U(1)_\psi$ rotates only the ψ_i . Show that each Abelian symmetry is anomalous and find a surviving linear combination. Calculate its $SU(p)^3$, $SU(p)^2U(1)$, $U(1)^3$, and mixed $U(1)$ –gravity 't Hooft anomalies.

The proposed confined spectrum consists of gauge-singlet fermions

$$\lambda_{ij} = \psi_{[i}(\chi\psi_{j]}), \quad (2.1)$$

in the antisymmetric representation of the global $SU(p)$. Find their surviving $U(1)$ charge and match all the anomalies computed above.

Useful source data: With $I(\bar{R}) = I(R)$ and $A(\bar{R}) = -A(R)$, the $SU(N)$ representation data are

R	fundamental	adjoint	symmetric	antisymmetric
$\dim R$	N	$N^2 - 1$	$N(N+1)/2$	$N(N-1)/2$
$I(R)$	1	$2N$	$N+2$	$N-2$
$A(R)$	1	0	$N+4$	$N-4$

Solution. For the theory to be consistent, we must have

$$\sum_i A(R_i) = A(\text{sym}) + pA(\bar{\mathbf{N}}) = N + 4 - p = 0 \implies p = N + 4. \quad (2.2)$$

Then, for the $U(1)_\chi SU(N)^2$ mixed anomaly, we have

$$\mathcal{A}_{U(1)_\chi SU(N)^2} = \sum q_i I(R_i) = I(\text{sym}) = N + 2, \quad (2.3)$$

while for the $U(1)_\psi SU(N)^2$

$$\mathcal{A}_{U(1)_\psi SU(N)^2} = \sum q_i I(R_i) = pI(\bar{\text{fund}}) = N + 4, \quad (2.4)$$

which are both anomalous. For a general $U(1)$ transformation, let us take the charge of χ to be Q and the charges of ψ_i to be q . For the symmetry to be anomaly-free, we must have

$$\mathcal{A}_{U(1) SU(N)^2} = pq + Q(N + 4) = 0. \quad (2.5)$$

A convenient choice of charge is $q = -(N + 2)$ and $Q = N + 4$.

Now, we turn to the computation of the 't Hooft anomalies:

$$\mathcal{A}_{SU(p)^3} = \sum_{\text{gauge index}} A(\mathbf{p}) = N, \quad (2.6)$$

$$\mathcal{A}_{SU(p)^2 U(1)} = \sum_{\text{gauge index}} qI(\mathbf{p}) = -N(N + 2), \quad (2.7)$$

$$\mathcal{A}_{U(1)^3} = \sum_{\text{gauge index}} \sum_i q_i^3 = Npq^3 + \dim(\text{sym})Q^3 = -\frac{N^3(N + 3)(N + 4)}{2}, \quad (2.8)$$

$$\mathcal{A}_{U(1)} = \sum_{\text{gauge index}} \sum_i q_i = Npq + \dim(\text{sym})Q = -\frac{N(N + 4)(N + 3)}{2}. \quad (2.9)$$

Under $U(1)$ the λ fermion transform as

$$U(1) : \lambda^{ij} \rightarrow e^{i(Q+2q)\alpha} \lambda^{ij}, \quad (2.10)$$

which has a charge $q_\lambda = -N$.

Since the λ fermion is a gauge singlet, the 't Hooft anomalies of this theory are given by

$$\mathcal{A}'_{SU(p)^3} = A(\text{antisym}) = p - 4 = N, \quad (2.11)$$

$$\mathcal{A}'_{SU(p)^2 U(1)} = q_\lambda I(\text{antisym}) = -N(p - 2) = -N(N + 2), \quad (2.12)$$

$$\mathcal{A}'_{U(1)^3} = \sum_{\text{fermion index}} q_\lambda^3 = -\frac{1}{2}p(p - 1)N^3 = -\frac{N^3(N + 3)(N + 4)}{2}, \quad (2.13)$$

$$\mathcal{A}'_{U(1)} = \sum_{\text{fermion index}} q_\lambda = -\frac{1}{2}p(p - 1)N = -\frac{N(N + 3)(N + 4)}{2}. \quad (2.14)$$

Therefore, we see that the anomalies indeed match.

3. Sheet 3, Problem 3*: Real Hypercharges and Anomaly Cancellation

Use one Standard Model generation, omitting a right-handed neutrino, with arbitrary real hypercharges:

$$Q_L : (\mathbf{2}, \mathbf{3})_q, \quad L_L : (\mathbf{2}, \mathbf{1})_l, \quad u_R : (\mathbf{1}, \mathbf{3})_u, \quad d_R : (\mathbf{1}, \mathbf{3})_d, \quad e_R : (\mathbf{1}, \mathbf{1})_x. \quad (3.1)$$

The entries specify $SU(2) \times SU(3)$ representations and the subscript is the $U(1)$ charge.

- i) Write every gauge anomaly-cancellation condition, including the mixed $U(1)$ -gravity condition.
- ii) Show that the real solutions consist of two branches: one with $u = -d$ and another with the usual Standard Model hypercharges up to overall rescaling.

Solution.

- i) We start with the $SU(3)^3$ gauge anomaly, which only involves the quarks:

$$\sum_{\text{left}} A(\mathbf{3}) = \sum_{\text{right}} A(\mathbf{3}), \quad (3.2)$$

which is indeed true. For $SU(2)^3$, we simply have $A(\mathbf{2}) = 0$, and hence is always anomaly-free. For $U(1)^3$, we have

$$\sum_{\text{left}} Y_i^3 = \sum_{\text{right}} Y_i^3 \implies 6q^3 + 2l^3 = 3u^3 + 3d^3 + x^3. \quad (3.3)$$

Now, we turn to the mixed gauge anomalies. For $SU(2)^2 U(1)$, which only involves left-handed particles, we have

$$\sum_{\text{left}} Y_i I(\mathbf{2}) = \sum_{\text{right}} Y_i I(\mathbf{2}) \implies 3q + l = 0. \quad (3.4)$$

For $SU(3)^2 U(1)$, we find

$$\sum_{\text{left}} Y_i I(\mathbf{3}) = \sum_{\text{right}} Y_i I(\mathbf{3}) \implies 2q = u + d. \quad (3.5)$$

Finally, $\text{grav}^2 U(1)$ gives

$$\sum_{\text{left}} Y_i = \sum_{\text{right}} Y_i \implies 6q + 2l = 3u + 3d + x. \quad (3.6)$$

- ii) The 4 equations boil down to

$$6u^3 + 6d^3 + 21u^2d + 21ud^2 = 0. \quad (3.7)$$

Clearly, this has the solution $u/d = -1 \equiv y$. The polynomial factorises to

$$(y + 1)(6y^2 + 15y + 6) = 0, \quad (3.8)$$

which has solutions, $y = -1, -2, -1/2$. Since u_R and d_R have the same representation, we can always relabel them. Therefore, we actually only have two possible classes of solutions:

$$u = -d, \quad q = 0, \quad l = 0, \quad x = 0, \quad (3.9)$$

$$u = -2d, \quad q = \frac{d}{2}, \quad l = \frac{3d}{2}, \quad x = 3d. \quad (3.10)$$

4. Sheet 3, Problem 4: Integer Hypercharges

Repeat Problem 3 with $q, l, u, d, x \in \mathbb{Z}$.

- i) Impose all gauge anomaly conditions, without imposing the mixed gravitational condition.
- ii) Show that the solution is unique up to overall scaling and automatically cancels the mixed gravitational anomaly. You may quote a familiar pure-mathematics result without proving it.

Hint: First write $u - d = 2y$ for integer y . Then use the variables v, w related by

$$x = -\frac{6}{v+w}, \quad y = \frac{3(v-w)}{v+w}. \quad (4.1)$$

Solution. Since $u + d = 2q \in 2\mathbb{Z}$, we must have $u - d = 2y \in 2\mathbb{Z}$. Substituting this and $3q = -l$ into the $U(1)^3$ anomaly-free condition, we find

$$48q^3 + 3(q+y)^3 + 3(q-y)^3 + x^3 = 0, \quad (4.2)$$

which simplifies to

$$54q^3 + 18qy^2 + x^3 = 0. \quad (4.3)$$

Absorbing q into x, y , so that $x, y \in \mathbb{Q}$ and

$$54 + 18y^2 + x^3 = 0. \quad (4.4)$$

Making the redefinition of variables and working through a bit of algebra, we find

$$v^3 + w^3 = 1, \quad (4.5)$$

which is the famous Fermat's last theorem with $N = 3$. This, famously enough, has only trivial solutions, namely $v = 1, w = 0$ or $v = 0, w = 1$. Again, these are equivalent solutions since we can always relabel u and d . It is not hard to see that this is indeed the hypercharge of our world up to a scaling, and hence free of gravitational anomaly.

5. Sheet 3, Problem 5: Global Symmetries of the Standard Model

Determine the global symmetries of the classical Standard Model Lagrangian in each case:

- i) there are no right-handed neutrinos;
- ii) right-handed neutrinos exist and every allowed relevant and marginal interaction is included.

For each case, reassess the result after accounting for anomalies.

Solution.

- i) Naively, there seems to be a $U(1)^5$ global symmetry, which is

$$U(1) : \quad Q_L \rightarrow e^{i\alpha} Q_L, \quad (5.1)$$

$$d_R \rightarrow e^{i\beta} d_R, \quad (5.2)$$

$$u_R \rightarrow e^{i\gamma} u_R, \quad (5.3)$$

$$L_L \rightarrow e^{i\delta} L_L, \quad (5.4)$$

$$e_R \rightarrow e^{i\lambda} e_R. \quad (5.5)$$

However, the Yukawa couplings impose additional restrictions on the charges, which are

$$\alpha = \beta, \quad \alpha = \gamma, \quad \delta = \lambda. \quad (5.6)$$

This reduces the global symmetry to $U(1)^2$. In particular, we choose this to be $U(1)_B \times U(1)_L$. After accounting for the anomalies, the mixed anomaly with $SU(2)$ and $SU(3)$ vanishes for both lepton and baryon number symmetry. However, the mixed anomaly with hypercharge survives:

$$\mathcal{A}_{U(1)_B U(1)^2} = -\frac{1}{2} = -\mathcal{A}_{U(1)_L U(1)^2}. \quad (5.7)$$

Therefore, the only surviving global symmetry in the quantum theory is $U(1)_B - U(1)_L$.

- ii) The situation is almost identical after including right-handed neutrinos, since they have no hypercharge.

6. Sheet 3, Problem 6: Surviving Discrete Symmetries

Part (a) is optional on the source sheet.

- a) Couple $SU(N)$ Yang–Mills theory to a single massless adjoint left-handed Weyl fermion λ . Explain quantum consistency. Although the theory has $\mathcal{N} = 1$ supersymmetry, that fact is not needed here. Classically $\lambda \mapsto e^{i\alpha}\lambda$ is a $U(1)$ symmetry. Follow the shift of the theta angle to show that only $\mathbb{Z}_{2N}$ survives quantum mechanically. If confinement produces $\langle \text{Tr } \lambda\lambda \rangle \sim \Lambda_{\text{QCD}}^3$, determine the action of $\mathbb{Z}_{2N}$ on this condensate and count the vacua.
- b) Now couple $SU(N)$, with $N > 2$, to one Dirac fermion in the two-index symmetric representation. Determine its classical and quantum global symmetries. Assuming confinement and a condensate $\langle \bar{\psi}\psi \rangle \sim \Lambda_{\text{QCD}}^3$, count the vacua. Repeat for the two-index antisymmetric representation. The representation data in Problem 2 may help.

Solution.

- a) Since $A(\text{adj}) = 0$, the theory is free of gauge anomaly, and hence consistent. For the $U(1)$ global symmetry, the ABJ anomaly is

$$\mathcal{A}_{U(1)SU(N)^2} = I(\text{adj}) = 2N. \quad (6.1)$$

Hence, the theta term transform as

$$\theta \rightarrow \theta + 2\alpha N. \quad (6.2)$$

Therefore, only a discrete $\mathbb{Z}_{2N}$ survives. Under the discrete symmetry, the condensate transforms as

$$\langle \text{Tr } \lambda\lambda \rangle \rightarrow e^{2in\pi/N} \langle \text{Tr } \lambda\lambda \rangle. \quad (6.3)$$

Therefore, the symmetry is further broken to $\mathbb{Z}_2$, which means that there are N vacuum states for this theory.

- b) In this case the classical global symmetry is $U(1)_V \times U(1)_A$, while the quantum global symmetry reduces to $U(1)_V$ and a surviving discrete group of $U(1)_A$. In particular, the theta angle transform as

$$\theta \rightarrow \theta + 2\beta(N \pm 2) \quad (6.4)$$

(+ for symmetric representation and $-$ for antisymmetric representation). Therefore, the surviving discrete symmetry is $\mathbb{Z}_{N \pm 2}$. Under the discrete symmetry, the condensate transforms as

$$\langle \bar{\psi}\psi \rangle \rightarrow e^{2\pi ni/(N \pm 2)} \langle \bar{\psi}\psi \rangle. \quad (6.5)$$

Therefore, there are $N \pm 2$ vacuum for the theory.